

HERE ARE THE BENCHMARK RESULTS

Benchmark	Ryzen AI 7 PRO 360	Core Ultra 7 155H	Core Ultra 7 256V	
Text Generation (LLM)	7.42	6.99	8.51	Tokens/sec (More is better)
Prompt Processing	13.91	33.78	22.31	Watts (Less is better)
Power Consumption	26.33	19.15	26.31	Tokens/sec (More is better)
Speech to Text (Whisper)	741.65	1320.32	624.47	Seconds (Less is better)
PyTorch Training Loop	37.15	19.78	34.87	Batches/sec (More is better)

tokens per second, both outperforming Intel’s Core Ultra 7 155H in equivalent workloads. This level of performance makes the system viable for casual research, summarisation tasks, note expansion, and even offline writing assistance—all happening offline.

More importantly, during sustained usage, the Ryzen AI 7 PRO 360 consumed less than 14 Watts, compared to over 33 Watts drawn by the Intel 155H and a little over 22 Watts on the Intel

mobile platforms featuring integrated NPUs, though with different performance and power envelopes.

We ran a suite of AI-focused benchmarks, designed to simulate typical on-device workloads. Here’s what we found.

Text Generation and Prompt Processing:

The Ryzen AI 7 PRO 360 outperformed the Core Ultra 7 155H by around 5–7% in token generation and by more than 36%

155H. This is critical in mobile contexts, where battery life and thermal load are significant constraints. Lower power draw also means more consistent performance without needing to throttle the NPU or CPU.

PyTorch Batching:

In machine learning training scenarios (even with modest batch sizes and local models), the Ryzen AI 7 PRO 360 achieved 37.15 batches per second, nearly double the performance of Intel’s chip. While most end users won’t be training models on laptops, this suggests AMD’s NPU and memory fabric can sustain higher throughput under repeated matrix workloads.

Whisper Transcription:

While AMD’s chip still leads in transcription time, the workload is more CPU/GPU-dependent, and not yet optimised for NPU acceleration on either platform. AMD still edges ahead with a nearly 40% faster transcription speed, reinforcing its position as a strong performer for offline productivity tools.

These numbers all but confirm that AMD’s Ryzen AI 7 PRO 360 is not just theoretically competitive—it delivers class-leading performance in several relevant on-device AI tasks, all while remaining extremely efficient on resources. For professionals seeking powerful inference capabilities without sacrificing mobility, the Ryzen AI 300 series sets a new baseline. **d**

256V in a similar task, highlighting the efficiency of AMD’s NPU for sustained AI inference.

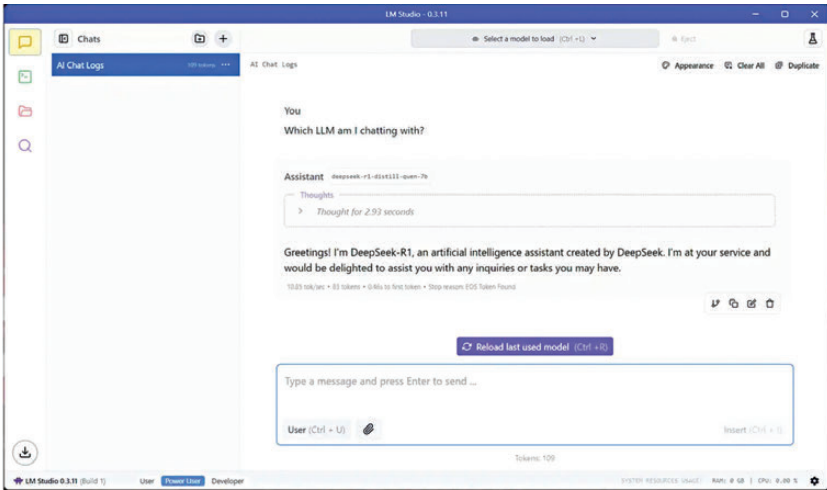
Benchmarks

While real-world applications tell one side of the story, benchmark testing provides a quantifiable view of how the Ryzen AI 7 PRO 360 stacks up against its closest competition—in this case, Intel’s Core Ultra 7 155H from the Meteor Lake family. Both chips represent their respective vendors’ latest

in prompt processing throughput. This is crucial for scenarios involving chat-based interfaces, autocomplete, summarisation, and structured inference tasks. AMD’s faster LPDDR5X memory (7500 MT/s) and broader cache bandwidth appear to contribute to this advantage.

Power Efficiency:

Where AMD’s solution truly shines is in efficiency. At 13.91 W of power consumption during AI inference, it drew less than half the power of the Core Ultra 7



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